VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
PRELIMINARY EXAMINATION 2009

PROTEOMICS **9815/01**
Higher 3 **28 September 2009**

Paper 1 **2 hours 30 minutes**

READ THESE INSTRUCTIONS FIRST

Write your name, NRIC and class on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.
Section B
Answer **three** out of **four** questions.
Section C
Answer the question.

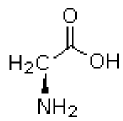
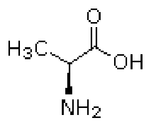
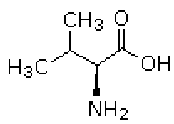
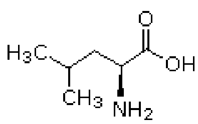
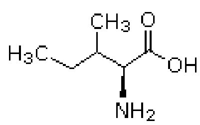
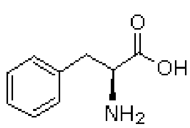
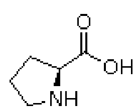
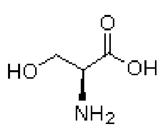
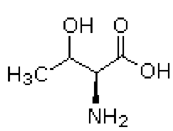
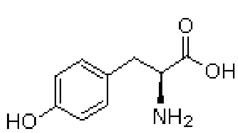
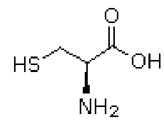
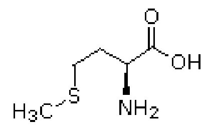
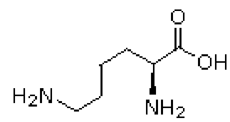
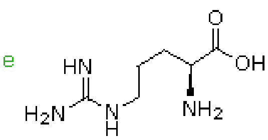
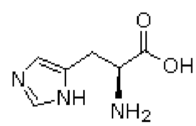
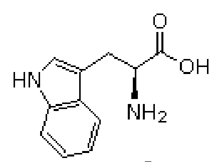
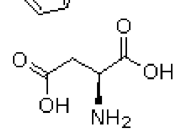
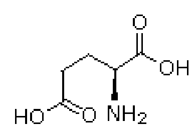
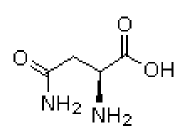
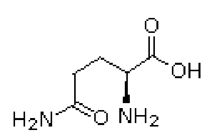
You may use a calculator.
You are reminded of the need for clear presentation in your answers.
At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

This document consists of **11** printed pages

Section A

Answer all questions in this section.

1. Below shows the 20 different amino acids that are most commonly found in proteins.

Name	Formula	Abbreviations	
Glycine		Gly	G
Alanine		Ala	A
Valine		Val	V
Leucine		Leu	L
Isoleucine		Ile	I
Phenylalanine		Phe	F
Proline		Pro	P
Serine		Ser	S
Threonine		Thr	T
Tyrosine		Tyr	Y
Cysteine		Cys	C
Methionine		Met	M
Lysine		Lys	K
Arginine		Arg	R
Histidine		His	H
Tryptophan		Trp	W
Aspartic Acid		Asp	D
Glutamic Acid		Glu	E
Asparagine		Asn	N
Glutamine		Gln	Q

(http://www.cem.msu.edu/~reusch/VirtualText/Images3/aminacid.gif)

Based on the list of amino acids, answer the following questions:

(a) "Nearly all amino acids have an asymmetric carbon atom." What is (are) the consequence(s) of this observation? [4]

(b) The amino acids of a polypeptide affect the shape of the protein. Predict where each of the following amino acid pairs will be found in the protein and explain why.

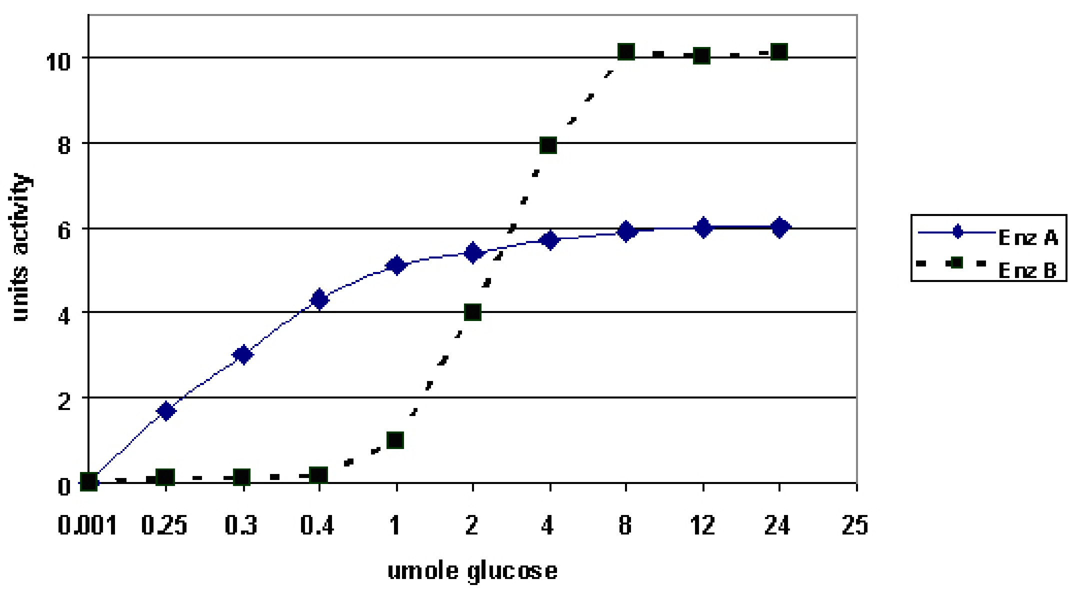
i. One amino acid is arginine and the other is serine. [1]
ii. One amino acid is alanine and the other is isoleucine. [1]

(c) "Most proteins incorporate small quantities of glycine." Explain why. [2]

(d) Sickle cell anemia results from a single base substitution. The resultant β globin chain, of haemoglobin S, has a valine residue instead of a glutamic acid. What do you expect the effect on the β globin chain might be if the mutation were to involve

a. leucine, instead of valine? [1]
b. aspartic acid, instead of valine? [1]

2. A researcher isolated two kinase enzymes (**A** and **B**) which catalyses the phosphorylation of **glucose** to **glucose-6-P** and **glucose-1-P** respectively. The rate of reactions in the presence of the same constant amount of enzyme **A** and enzyme **B** were measured with varying amounts of **glucose** in two experiments. The results are plotted as shown below:



(a) Explain which enzyme, **A** or **B**, has the greater affinity for its own substrate? [3]

(b) If enzyme **A** were subjected to non-competitive inhibition by the drug insulineride, explain what you would you expect the Vmax to be? [2]

(c) From the plot shown by enzyme **B,** explain the advantage of having enzyme **B** in a cell. [2]

(d) The researcher further found that for enzyme **A**, ATP participates in the reaction in a form complexed to magnesium (Mg) as a cofactor. What is a cofactor and explain the role of magnesium as a co-factor in enzyme **A** activity. [3]

3. The study of the regulation of the expression of luteinizing hormone (LH) β gene in mammalian pituitary cells involves identifying transcription factors and related co-factors during its transcription. The αT3 and LβT2 mouse pituitary cells represent the early and fully differentiated pituitary cells respectively, where only the latter synthesizes LHβ protein. Thus, the nuclear proteins uniquely expressed in the LβT2 cells were postulated to be responsible for LHβ gene transcription, and this was investigated by comparing the proteome of two cell types using two-dimensional gel electrophoresis.

(a) Explain what is meant by the term "proteome". [1]

(b) Outline the steps carried out to obtain the proteome of the αT3 and LβT2 mouse pituitary cells. [4]

The initial results of the 2D gels were shown in figure 1 and 2.

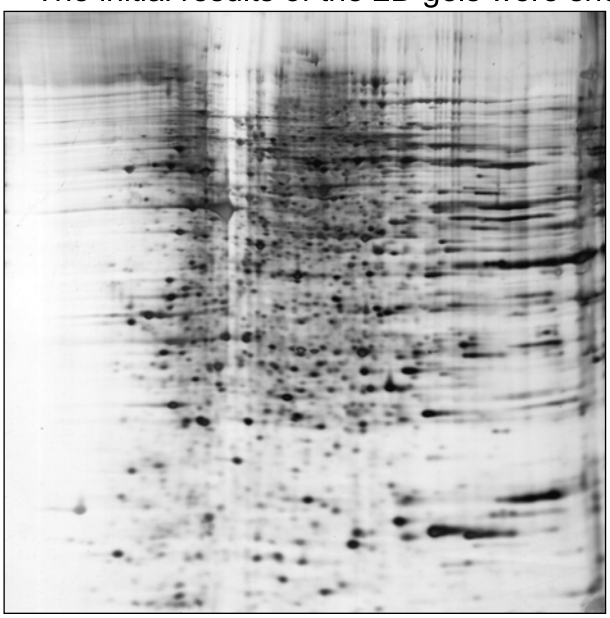


Figure 1: αT3 cell profile

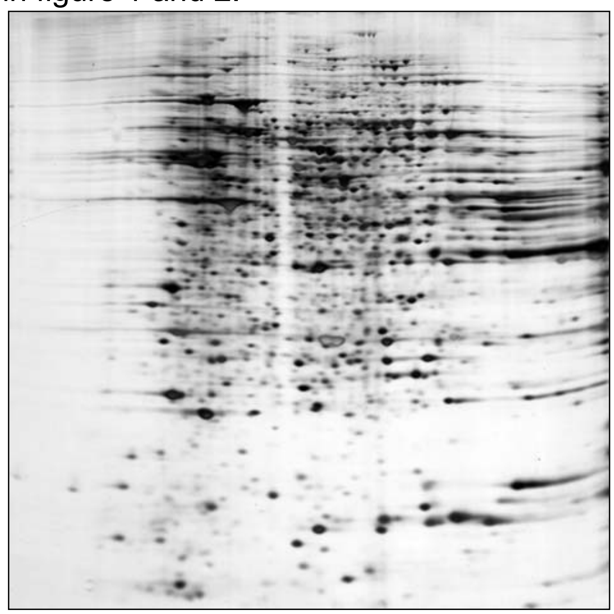


Figure 2: LβT2 cell profile

(c) With reference to figure 1 and 2, explain what can be done to improve the resolution of the gel and subsequent identification of nuclear proteins [1]

The improved results of the 2D gels are now shown in figure 3 and 4.

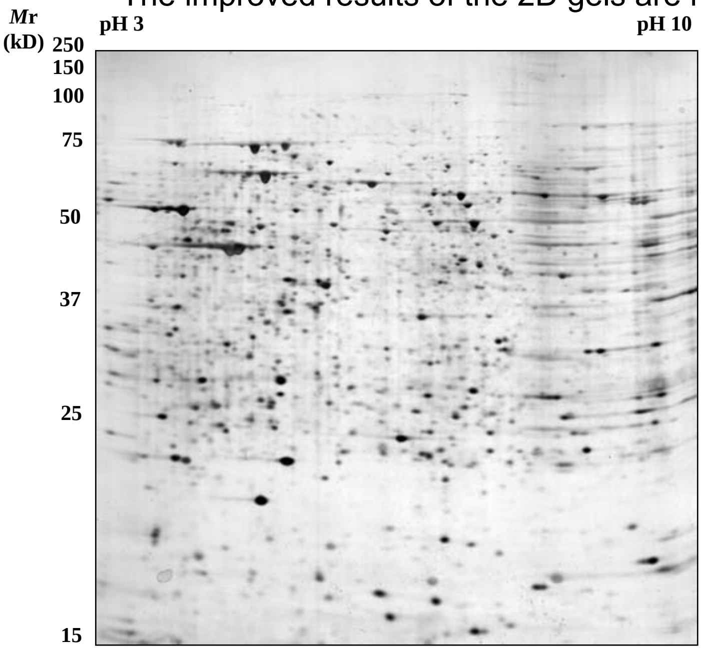


Figure 3: αT3 cell profile

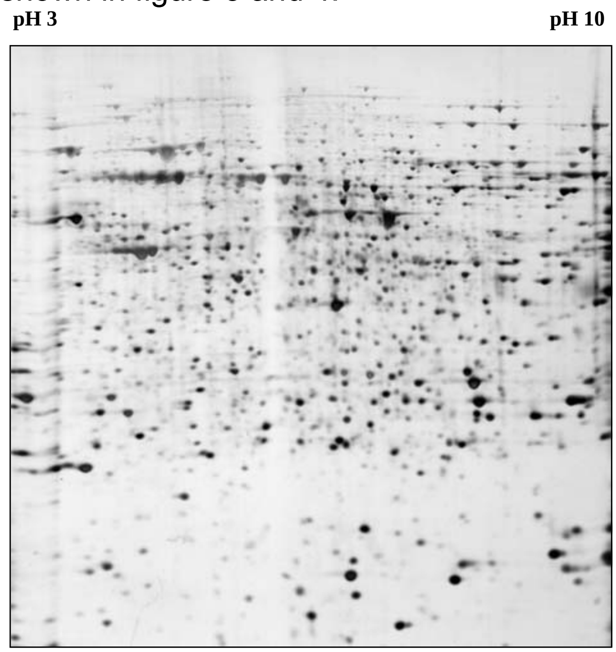
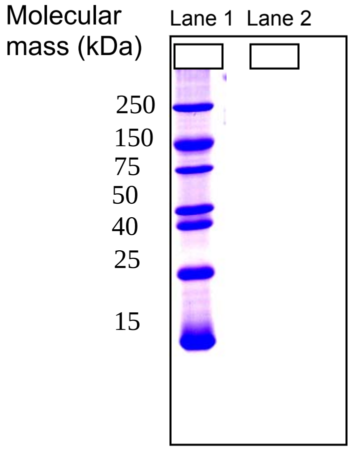


Figure 4: LβT2 cell profile

(d) On figure 4, circle one possible protein spot on the that may be a nuclear protein [1]

(b) When the amino acid sequence of the protein was determined, the coding sequence of the gene was determined and cloned into an expression vector. The recombinant expression vector was introduced into *E.coli* cells to synthesize large quantity of the protein for further study. The purified protein was loaded onto a SDS-PAGE gel

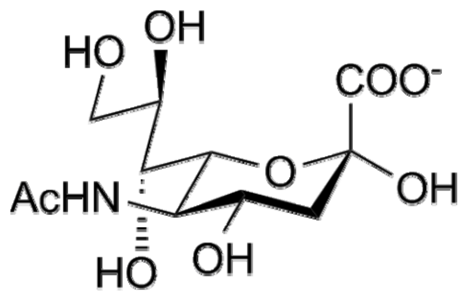
(i) On the diagram below, indicate the result of SDS-PAGE in lane 2. [1]



Legend
Lane 1: Protein ladder

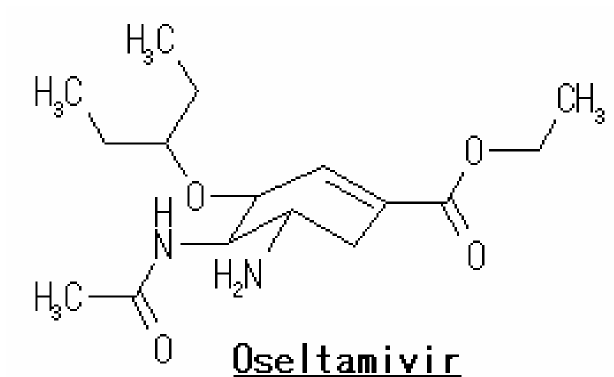
(ii) Describe one method that may help you identify the correct protein band in lane 2. [2]

4. Influenza A viruses are classified in subtypes according to the kind of antibodies that these proteins raise. Sixteen subtypes of H (HA) and nine types of N (NA) are known. The Influenza virus H1N1 is then the Influenza A virus with Hemagglutinin subtype H1 and Neuraminidase subtype N1.
Once the viruses have been replicated inside the cell, they are excreted from the host cell inside a spherical phospholipid membrane that contains Hemagglutinin and Neuraminidase. In the same way that happened at the beginning of the infection, the Hemagglutinin binds to sialic acid residues, but the Neuraminidase hydrolyses the glycoside linkage between Hemagglutinin and the Sialic Acid residues.



N-Acetyl-Neuraminic Acid (Sialic acid)

The drugs Oseltamivir (Tamiflu®) and Zanamivir (Relenza ®) have been effective in the treatment of infections with H1N1 virus. The structure of Osltamivir is shown below



Oseltamivir

(a) Suggest how Oseltamivir works to halt the proliferation of the H1N1 virus. [1]

(b) In the development of anti-viral drugs, it is often important to employ high throughput methods to determine binding partners to a query protein. State one such method. [1]

(c) Explain how RNA interference can be used to characterize the cellular roles of proteins. [3]

(d) Explain why RNA interference is not the method of choice to elucidate the cellular roles of membrane proteins on the viral envelope. [1]

(e) Discuss four characteristics of an efficient high throughput cloning system. [4]

5. The binding of platelet-derived growth factor (PDGF) to the PDGF receptor stimulates the phosphorylation of 8 tyrosines in the receptor's cytoplasmic domain.

The enzyme PI 3-kinase binds to one or more of the phosphotyrosines through its SH2 domains and is thereby activated. To identify the activating phosphotyrosines, you synthesize 8 pentapeptides that contain the critical tyrosines (at the N-terminus) in their phosphorylated and unphosphorylated state.

You then mix an *excess* of each of the various pentapeptides with phosphorylated PDGF receptor and PI 3-kinase. Immunoprecipitation of the PDGF receptor will bring down any bound PI 3-kinase, which can then be assayed by its ability to add ^{32}P-phosphate to its substrate (Figure 5.1).

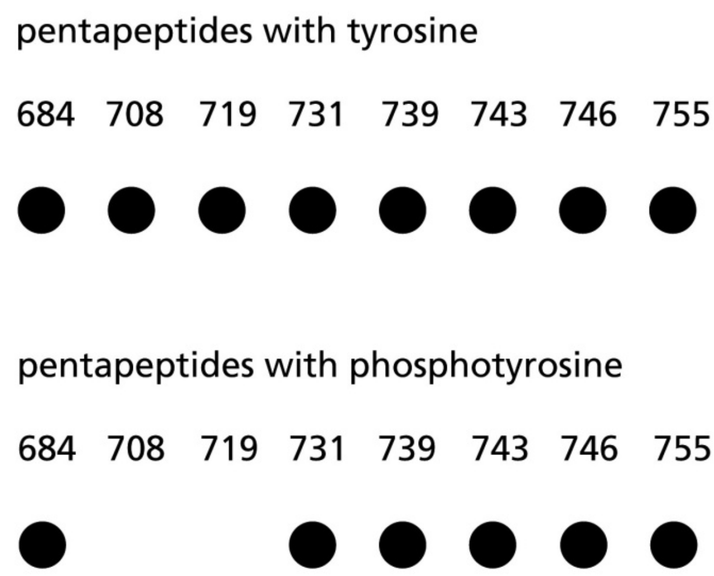


Figure 3-24 MBoC5: The Problems Book (© Garland Science 2008)

Figure 5.1

(NB: the number indicates the position of the pentapeptides in the PDGF receptor. The dark circles indicates the incorporation of ^{32}P-phosphate into the substrate by PI 3-kinase)

(a) What do you understand by the term "domain"? [2]

(b) Explain the results shown in Figure 5.1 with regard to the binding of PI 3-kinase to phosphotyrosines in the activated PDGF receptor. [3]

The amino acid sequences of the PDGF receptor pentapeptides tested above and the peptide segments of some other receptors that are known to bind to PI 3-kinase are shown in **Figure 5.2**.

684	YSNAL		YMMMR	(FGF receptor)
708	YMDMS		YTHMN	(insulin receptor)
719	YVPML		YEVML	(hepatocyte growth factor receptor)
731	YADIE	*PDGF*	YMDMK	(steel factor receptor)
739	YMAPY		YVEMR	(CSF-1 receptor)
743	YDNYE			
746	YEPSA			
755	YRATL			

Figure 5.2

(c) What are the common features of peptide segments that form the binding sites for PI 3-kinase? [2]

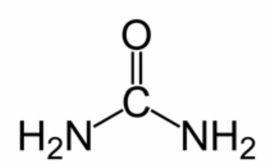
(d) Name the type of protein-protein interaction between PI 3-kinase with the PDGF receptor. Explain your answer. [2]

(e) State another type of protein-protein interaction. [1]

Section B

Answer 3 out of the 4 questions in this section.

6. The structure of urea is shown below:



(http://www.interatlaschemical.com/UREA/Urea%20molecular%20structure.png)

(a) High concentrations of urea will unfold proteins. Explain why. [2]

(b) Consider the following protein sequence: (Reference can be made to the list of amino acids of Question 1 in Section A)

Leu-Lys-Val-Asp-Ile-Ser-Leu-Arg-Leu-Lys-Ile-Arg-Phe-Glu

(i) What is the arrangement of amino acids in this sequence when incorporated into a β sheet? [2]

(ii) Predict how the β sheet, with the incorporated sequence of amino acids, may be arranged in a protein. [1]

(c) Consider a polypeptide composed entirely of glycine, i.e. polyglycine. Which environment, i.e. a hydrophobic or an aqueous environment, will be more likely to promote α helix formation in this polypeptide? Why? [3]

(d) A scientist has determined the sequence of an unknown protein. Using your knowledge of Bioinformatics, how will you predict the protein structure? [2]

7. You have isolated a novel protein from a newly discovered microorganism that lives in thermal vents near Mt. Pinatubo, Philippines. The genomic sequence is unknown and unavailable. However you have managed to sequence its amino acid sequence through the Edman degradation method.

(a) Describe a biological database. [3]

(b) Describe an example of a biological database that can be used to deduce a possible function of the novel protein and one kind of information that can be obtained from it. [3]

The protein was later found to contain a binding domain similar to transcription factors.

(c) Describe briefly how the protein was found to contain the domain. [2]

(d) The microorganism was later found to carry a genome that is smaller than *E.coli*. Surprisingly it has almost the same number of functional proteins. Suggest a reason for this. [2]

8. The characterization of protein modifications is essential for the study of protein function using functional genomic and proteomic approaches. technique to determine protein phosphorylation sites as well as the sequences and several previously unknown modifications of 28 small proteins isolated from *Escherichia coli* K12 cells.

(a) Like other forms of posttranslational modification, the covalent attachment of phosphate groups to protein targets at any of a number of surface Asp, His, Ser, Thr, or Tyr residues can profoundly affect protein behavior. State where such phosporylation is often needed. [3]

(b) In bacteria, secreted and extracellularly oriented membrane proteins are often stabilized by disulfide bonds, i.e., covalent links between the sulfhydryl groups of Cys residues in the same or different polypeptide chains. Suggest a role for such bonding. [1]

(c) The protein spot from a SDS-Page gel was cut and trypsin digested for identification using mass spectrometry. However, the protein databases were unable to identify the protein accurately. Explain how the method of Liquid chromatography coupled to mass spectrometry may determine the amino acid sequence of the unknown protein. [2]

(d) Explain how Edman degradation can also be used to determine the amino acid sequence of a polypeptide [2]

9. Smallpox, a devastating infectious disease dreaded throughout much of recorded history, is caused by variola virus, a member of the *Poxviridae* family of DNA viruses that replicate entirely in the cytoplasm. However, the mechanisms that contribute to smallpox pathogenesis in man are poorly understood at the molecular level. Recently scientists have conducted proteomic screening of the variola virus by using yeast-2-hybrid screening against a variety of human cDNA libraries.

(a) Describe how the yeast 2-hybrid is used to study protein-protein interactions in the above example. [4]

(b) Explain why a variety of cDNA library is used in this study. [2]

In the study, several protein-protein interactions were identified, including an interaction between variola G1R and the human nuclear factor NF-kappaB1/p105. NF-kappaB1/p105 acts as a transcription factor and plays and an important role in the inducible expression of genes involved in immune and inflammatory responses, apoptosis and development. The viral G1R protein blocks NF-kappaB signaling in human cells. This NF-kappaB inhibitor is found to be highly conserved among pathogenic orthopoxviruses.

(c) Explain how the variola G1R can cause death in humans. [2]

(d) Explain what it meant by the G1R protein being highly conserved and the significance of this conservation to pathogenic orthopoxviruses. [2]

Section C

Answer the question in this section

10 (a) Describe how bioinformatics can be used to analyse a novel protein. [10]

(b) Suggest how amino acid sequences similarity may provide clues about protein functions. [5]

(c) Describe how sequence homology searches can be useful in identifying close relatives. [3]

(d) Identify the benefits of using two sequence comparison softwares simultaneously. [2]